MACHINE LEARNING · NLP · MENTAL HEALTH

Depression Detection from Tweets

Text Classification with TF-IDF and Multiple Classifiers

Dataset: 123,759 tweets · Dec 2019 – Dec 2020 · Binary Classification: Depressive vs Non-Depressive

PROBLEM & DATASET

The Problem

- Depression is a leading cause of global disability.
- Social media = rich source of emotional signals.
- Goal: Automatically detect depressive tweets.
- Period: COVID-19 pandemic amplified mental health discussions online.

Dataset

123,759

cleaned tweets

48.7%

depressive (0)

51.3%

non-depressive (1)

PREPROCESSING PIPELINE

01

Data Cleaning

Remove URLs, @mentions, hashtag symbols, RT tags. Drop 10,314 duplicates & NaN rows.

02

Lemmatization

POS tagging + WordNet lemmatizer. Remove stop words & tokens < 3 chars.

03

Exploratory Data Analysis

Word frequencies, TF-IDF ranking, unigrams, bigrams, trigrams per class.

04

Ratio Filter

Remove 1,656 neutral words (ratio 0.35–0.65, freq ≥ 50) from both classes.

05

Term Frequency – Inverse Document Frequency.

Custom stopwords applied. Max 5K–10K features. Sublinear TF scaling.

★ *Novel contribution: ratio-based neutral word filtering*

KEY EDA FINDINGS

Top Depressive (0) Words

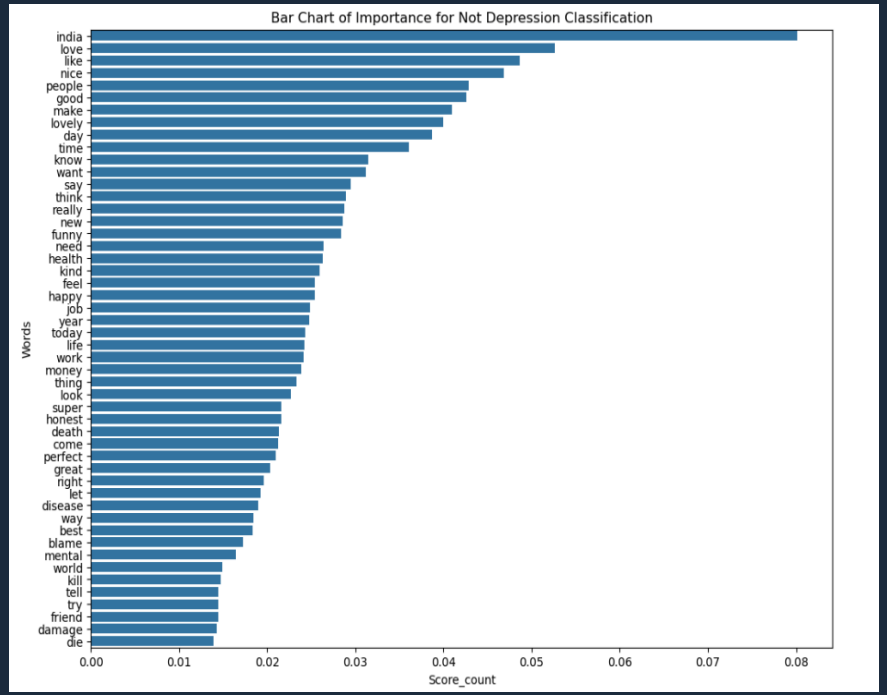
sad awful anger
broken alone anxiety
horrible hate kill
bad difficult blame
break shit

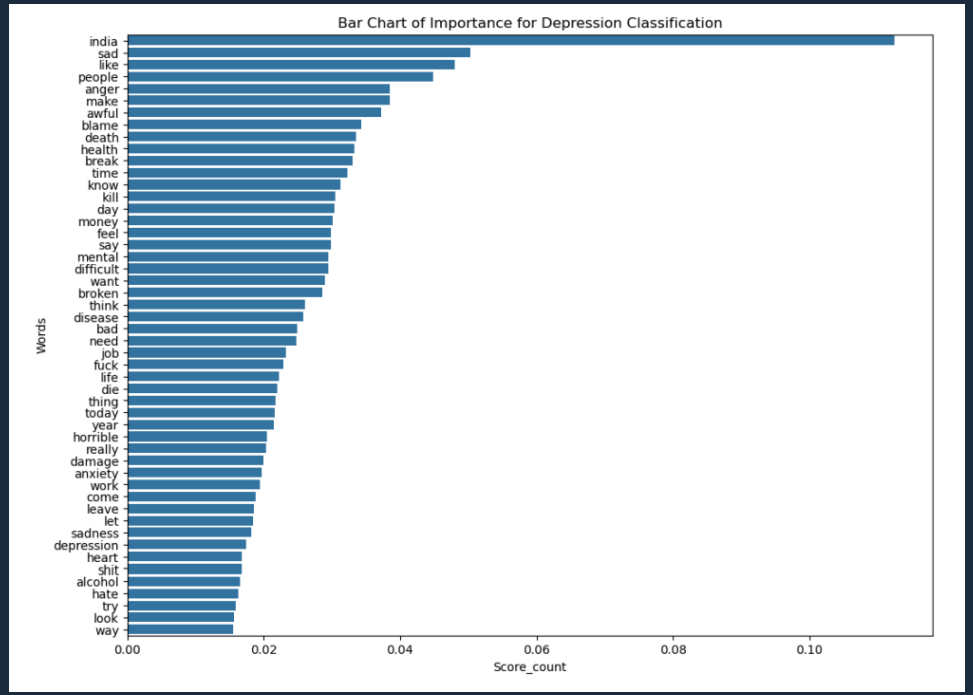
Top Non-Depressive (1) Words

nice love good
lovely funny kind
happy perfect honest
super more new
much many most

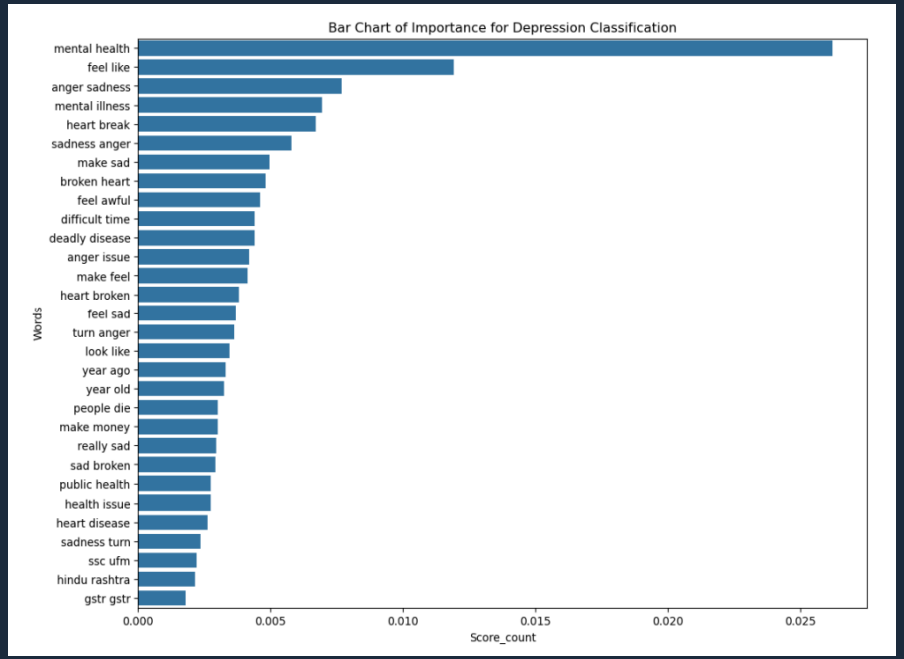
Tweet length (words/chars) shows NO significant difference between classes → classification relies entirely on lexical semantics

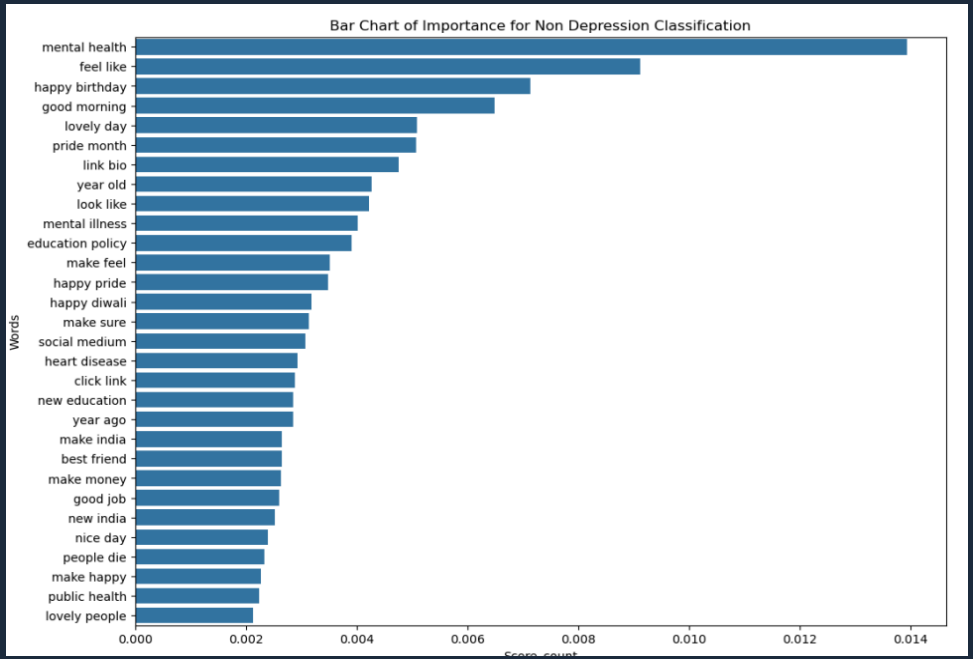
Unigrams


Bar Chart of Importance for Not Depression Classification
india
love
like
nice
people
good
make
lovely
day
time
know
want
say
think
really
new
funny
need
health
kind
feel
happy
job
year
today
life
work
money
thing
look
super
honest
death
come
perfect
great
right
let
disease
way
best
blame
mental
world
kill
tell
try
friend
damage
die
Words
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
Score_count

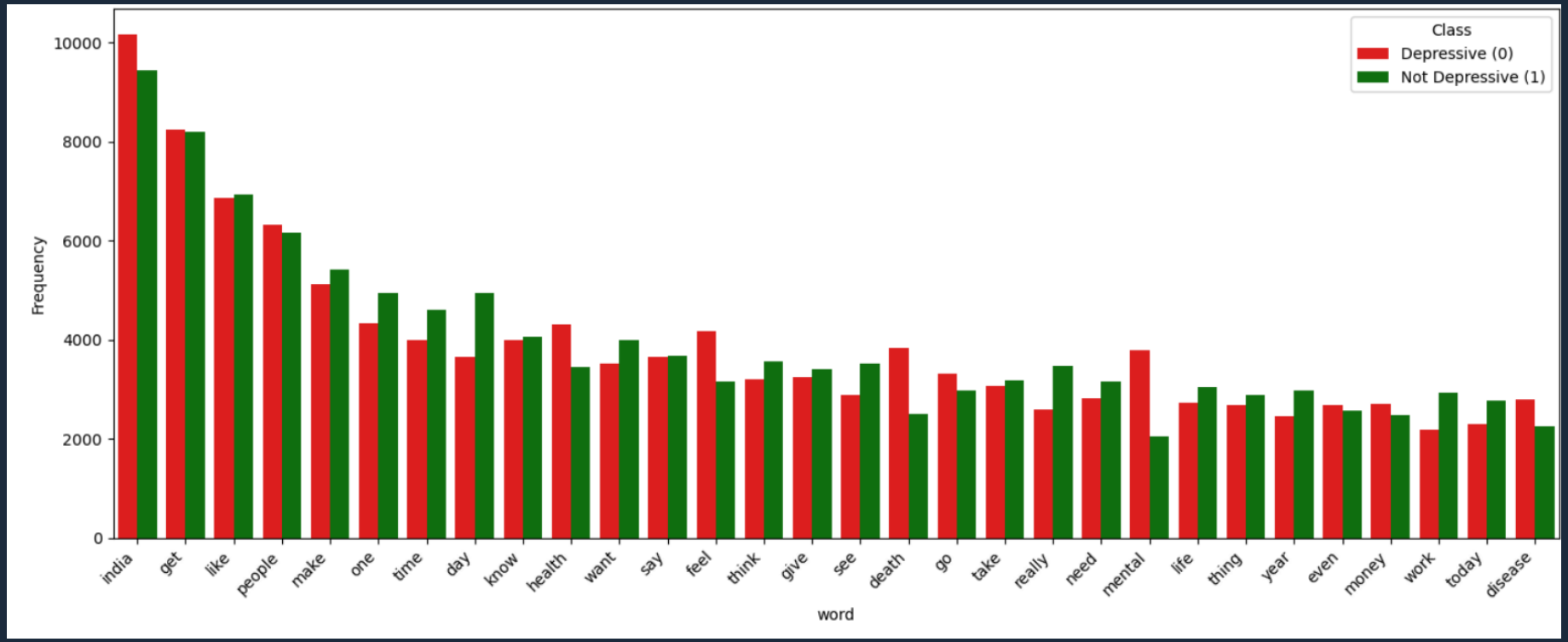

Bar Chart of Importance for Depression Classification
india
sad
like
people
anger
make
awful
blame
death
health
break
time
know
kill
day
money
feel
say
mental
difficult
want
broken
think
disease
bad
need
job
fuck
life
die
thing
today
year
horrible
really
damage
anxiety
work
come
leave
let
sadness
depression
heart
shit
alcohol
hate
try
look
way
Words
0.00
0.02
0.04
0.06
0.08
0.10
Score_count

Bigrams


Bar Chart of Importance for Depression Classification
mental health
feel like
anger sadness
mental illness
heart break
sadness anger
make sad
broken heart
feel awful
difficult time
deadly disease
anger issue
make feel
heart broken
feel sad
turn anger
look like
year ago
year old
people die
make money
really sad
sad broken
public health
health issue
heart disease
sadness turn
ssc ufm
hindu rashtra
gstr gstr
Words
0.000
0.005
0.010
0.015
0.020
0.025
Score_count


Bar Chart of Importance for Non Depression Classification
mental health
feel like
happy birthday
good morning
lovely day
pride month
link bio
year old
look like
mental illness
education policy
make feel
happy pride
happy diwali
make sure
social medium
heart disease
click link
new education
year ago
make india
best friend
make money
good job
new india
nice day
people die
make happy
public health
lovely people
Words
0.000
0.002
0.004
0.006
0.008
0.010
0.012
0.014

Top 20 neutral words



MODELS & METHODOLOGY

Feature Extraction

- TF-IDF vectorization

Transforms text into a numerical matrix, where each document is represented by the weighted importance of its words

- Sublinear TF scaling

using log(1 + TF) to reduce the effect of very frequent words while keeping meaningful differences

- n-gram: (1,1) or (1,2)
- Max features: 5K–10K

***5K–10K** refers to **limiting the number of words (or tokens) the model uses** when creating the TF-IDF vectors*

- Custom stopwords:

Example: "the", "a", "in", "on"

- NLTK + 1,656 neutral

standard English stopwords + extra dataset-specific neutral words removed to improve text processing.

- ratio-filtered words

common words that appear roughly equally in all classes and are removed because they are uninformative.

- 80% of data for training, 20% for testing

Logistic Regression

max_iter=1000

Unigrams+Bigrams

10K features

Random Forest

100 trees

max_depth=25

n_jobs=-1

Linear SVM

C=1.0

max_iter=2000

Unigrams+Bigrams

XGBoost

n_estimators=100,

max_depth=3,

learning_rate=0.1

Naive Bayes (Uni+Bigrams)

MultinomialNB

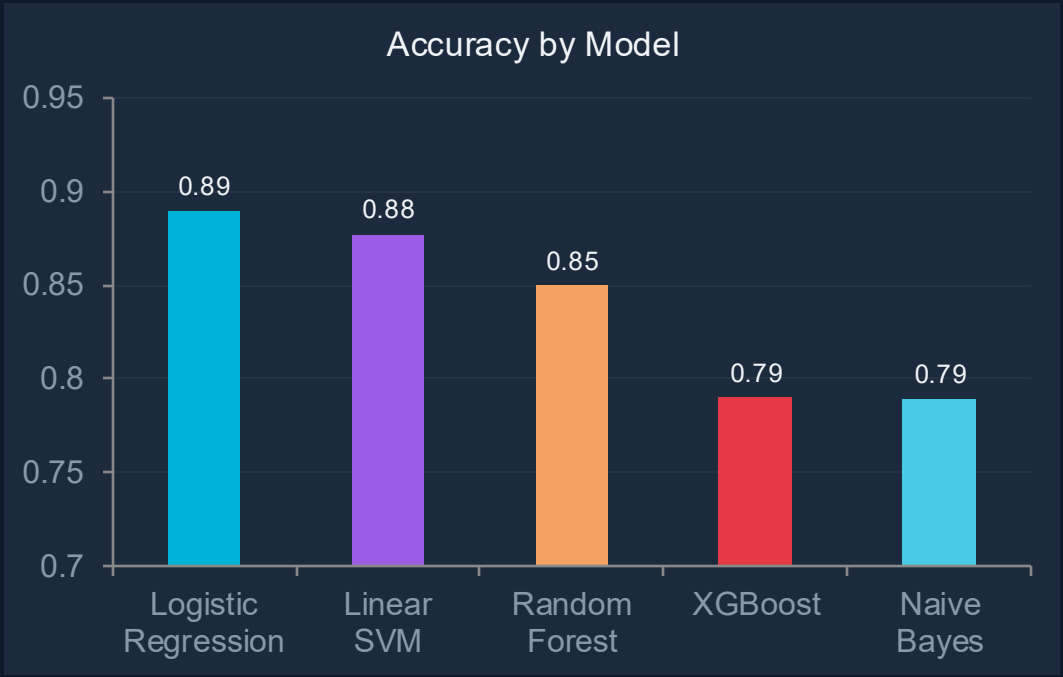
5K features

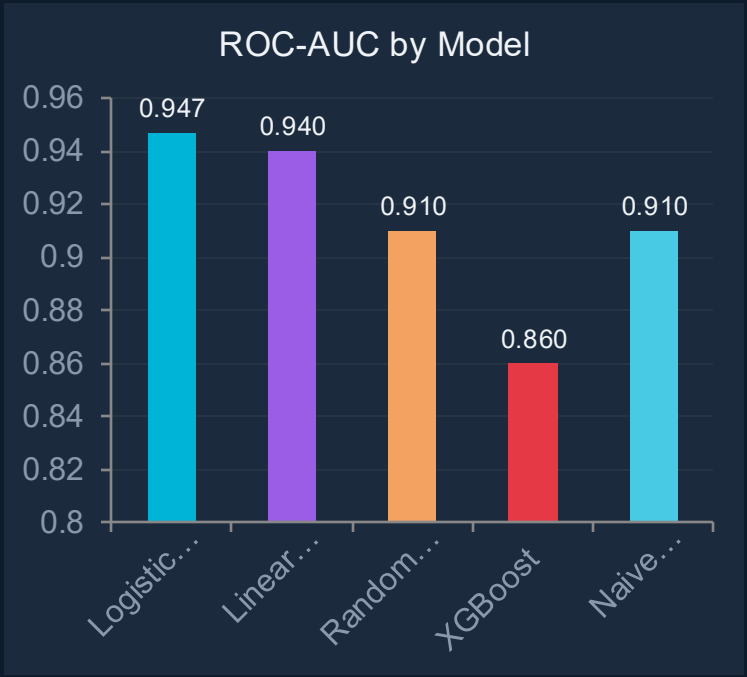
Sublinear TF

Evaluation

- Accuracy
- Precision / Recall / F1
- ROC-AUC
- Confusion Matrix

RESULTS & EVALUATION





Best Accuracy: Logistic Regression — 89.0%

NB AUC 0.90–0.91: Strong ranking despite lower accuracy

All models far exceed random baseline (AUC 0.50)

CONCLUSION

- Logistic Regression is best overall: 89.0% accuracy, AUC 0.947.
- Linear SVM a close second: 87.7%, AUC 0.94.
- Naive Bayes achieves AUC 0.90–0.91 — excellent ranking despite lower accuracy.
- Ratio-based neutral word filter (1,656 words) improved all models.
- Tweet length is NOT a discriminative feature — semantics alone drive classification.

Thank you for your time!